

PROJECT SHOWCASE #03

AIRLINE PASSENGER FORECASTING

Using Exponential Smoothing

Comparative Analysis of Single & Double Exponential Smoothing (2014 – 2024)



R



BPS



Time Series



Forecast Pkg



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Why Passenger Forecasting Matters?



Air Transportation

*Passenger demand analytics
for smarter operations*



Passenger Demand

Fluctuating demand requires accurate predictions for resource allocation.



Airline Capacity

Airlines need to optimize fleet capacity based on demand forecasts.



Airport Operations

Airports must prepare facilities ahead of peak passenger periods.



Decision Making

Data-driven forecasts enable better strategic planning decisions.



2014



Growth



COVID-19 (2020)



Recovery (2022+)



Forecast 2025

Dataset Overview

Monthly airline passenger departures
BPS Indonesia

SOURCE
BPS (Badan Pusat Statistik)

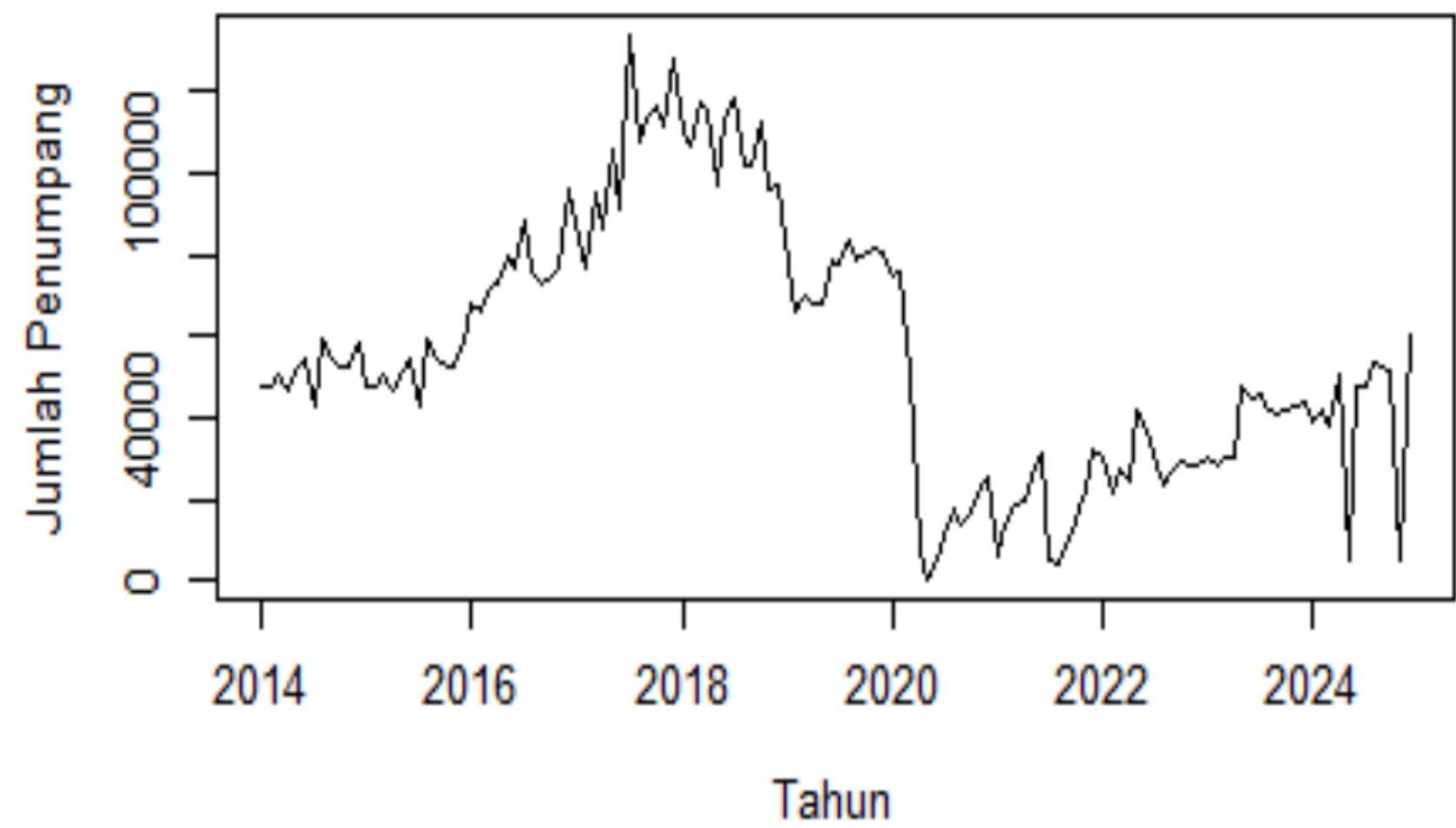
PERIOD
2014 – 2024

FREQUENCY
Monthly

OBSERVATIONS
132 Records

FORECAST HORIZON
12 Months (2025)

Data Deret Waktu Jumlah Penumpang Pesawat (2014-2024)



● COVID-19 Drop (2020)

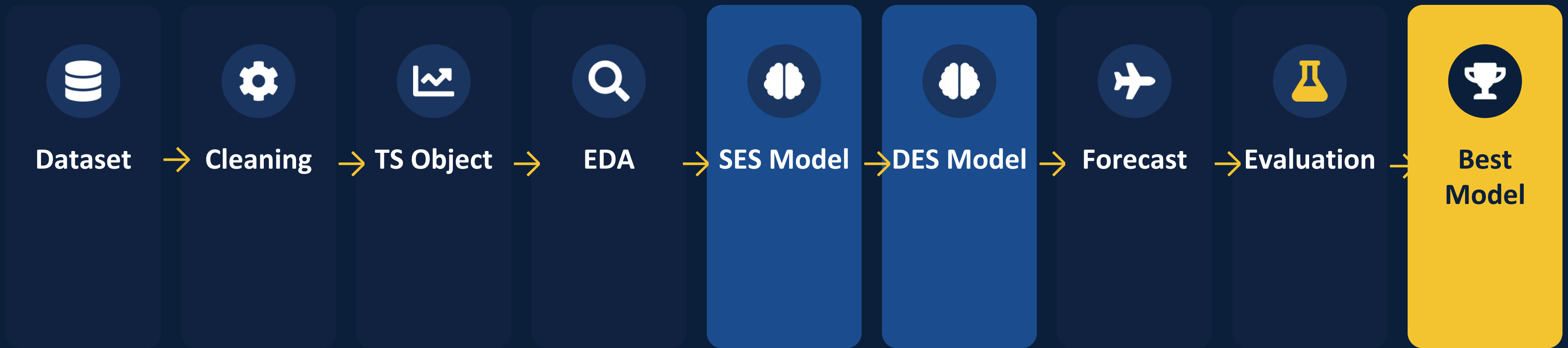
● Recovery (2022+)

INSIGHT

Data shows a strong upward trend (2014–2019), a severe crash due to COVID-19 (2020), and a gradual recovery post-2022 — confirming the need for trend-aware forecasting.

Forecasting Pipeline

End-to-end time series forecasting workflow in R



TOOLS  R Language  forecast  ggplot2  readr / dplyr  BPS Data

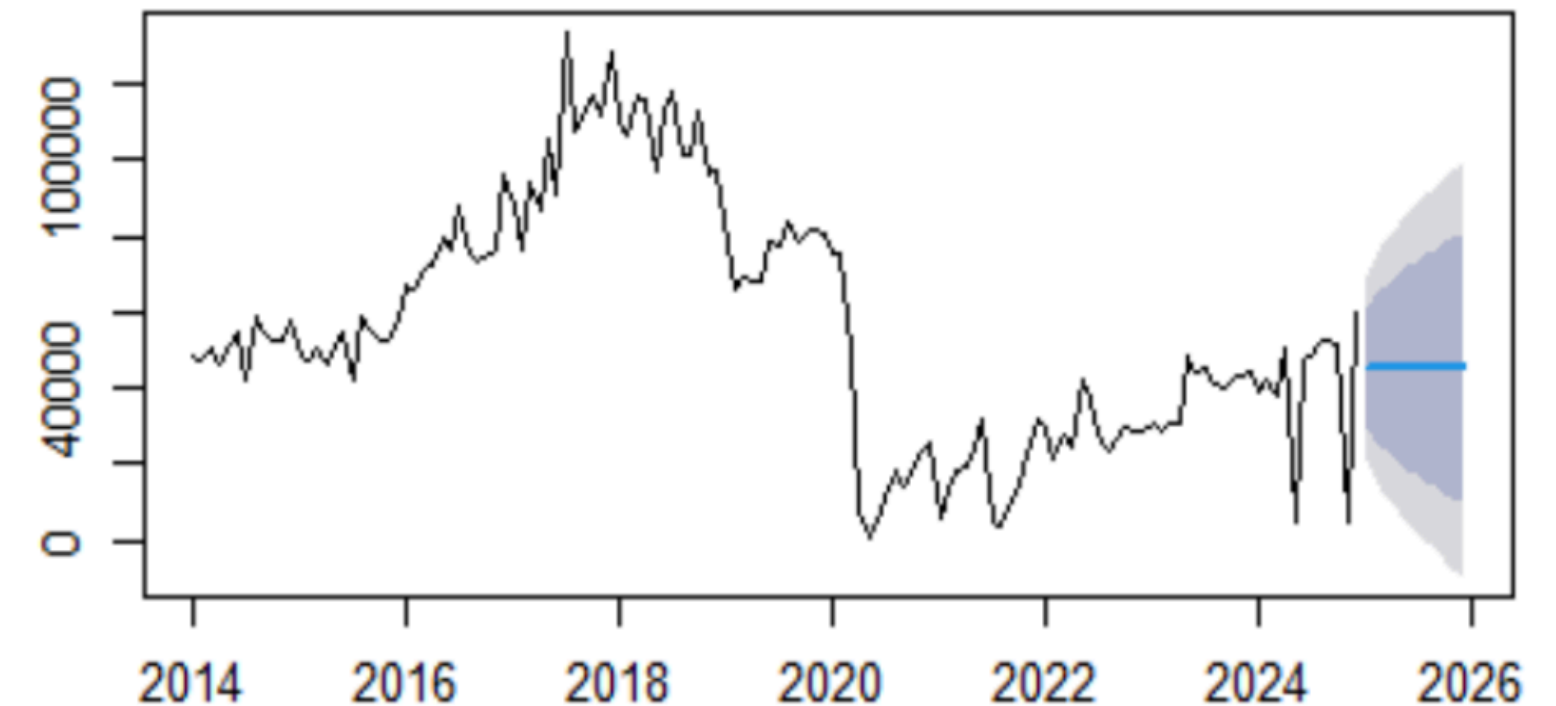
Forecasting Model Comparison

Single Exponential Smoothing (SES)



- Simple model — easy to implement
- Constant flat forecast line
- ⚠ Does NOT capture trend component

Double Exponential Smoothing (DES / Holt)



- Adaptive — captures trend
- Dynamic, non-flat forecast
- More representative of real data

COMPARISON

SES produces a constant horizontal forecast line. DES (Holt) follows the downward trend slope in recent data, producing a more realistic and adaptive projection.

2025 Forecast Result

Periode	Prediksi (Forecast)	Rentang bawah (95%)	Rentang atas (95%)
Januari 2025	45426	21174	69678
Februari 2025	45406	17027	73785
...	...	...	...
November 2025	45226	-7344	97795
Desember 2025	45206	-9398	99809

FORECAST TABLE (2025)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
SES:	45,459	45,459	45,459	45,459	45,459	45,459	45,459	45,459	45,459	45,459	45,459	45,459
DES:	45,426	45,406	45,386	45,366	45,346	45,326	45,306	45,286	45,266	45,246	45,226	45,206

SES FORECAST
45,459
Constant every month

DES FORECAST
45,426→45,206
Gradual decline (Jan→Dec)

DIFFERENCE
≈ 253
passengers (Dec 2025)

Performance Evaluation



MSE

Mean Squared Error

Average squared deviation — penalises large errors heavily.

✓ **SES: 148,444,051**



RMSE

Root Mean Squared Error

Square root of MSE; same unit as original data.

✓ **SES: 12,183.77**



MAPE

Mean Abs. Percentage Error

Error expressed as percentage — easy to interpret.

✓ **DES: 73.89%**

ACCURACY COMPARISON

Metric

SES

DES / Holt

Winner

MSE 148,444,051
RMSE 12,183.77
MAPE 73.95%

148,467,645
12,184.73
73.89%

SES ✓
SES ✓
DES ✓

BEST PRACTICAL MODEL



**Double Exponential
Smoothing (DES / Holt)**

Captures trend component



TREND DETECTED

2014–2019

Passenger data shows a clear upward trend across the observation period.



COVID-19 SHOCK

Min: 618

Pandemic caused a catastrophic drop in 2020 — the lowest recorded figure.



RECOVERY PHASE

2022+

Gradual recovery from 2022 with a consistent upward movement pattern.



DES MORE ADAPTIVE

73.89%

DES MAPE is slightly lower, capturing trend dynamics SES misses.



SES LOWER ERROR

MSE/RMSE

SES has marginally lower absolute error — both methods are closely matched.



BUSINESS VALUE

Actionable

Forecasts can guide airport capacity and airline fleet planning for 2025.

Let's Connect



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TECH STACK



R Language



BPS Dataset



forecast pkg



ggplot2



Time Series

Thank you for exploring this project.